



Corporal Punishment and Child Development in Low- and Middle-Income Countries: Progress, Challenges, and Directions

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Abstract

Most studies and reviews of studies on the developmental consequences of corporal punishment have focused on samples from the U.S. and other high-income countries. This study conducted a rapid review of the literature on the associations between corporal punishment and children's cognitive and social-emotional development in low- and middle-income countries (LMICs). Information from more than 42 studies of children younger than 18 years living in 64 LMICs was reviewed. Overall, the reviewed studies show associations between corporal punishment and negative cognitive and social-emotional outcomes, and there is no evidence that corporal punishment may relate to any positive developmental outcome in LMICs. Yet, issues of internal and external validity are common in the literature. The current evidence indicates that corporal punishment might increase the risk of detrimental child outcomes in LMICs, but further research with stronger methodological designs including samples from multiple settings is warranted.

Keywords Corporal punishment · Child development · Low- and middle-income countries

Two out of three children younger than five living in low- and middle-income countries (LMICs) are exposed to corporal punishment in the home [1]. Growing evidence from cross-sectional and longitudinal studies—mostly from the U.S and other high-income countries (HICs)—has shown that corporal punishment relates to an array of detrimental child outcomes, including internalizing and externalizing behavior problems and poorer cognitive abilities [2–4]. In response, the United Nations (UN) clarified that corporal punishment is a form of violence against children and recommended its eradication in the Convention on the Rights of the Child—CRC [5] and the Sustainable Development Goals—SDGs [6].

While considerations of children's rights are enough to recommend the eradication of all forms of corporal punishment, some scholars have criticized the global movement against corporal punishment arguing that the existing evidence on the association between corporal punishment and

child developmental outcomes is not conclusive, particularly in LMICs [7–9]. Assuming that there is scarce evidence on the links between corporal punishment and child developmental outcomes in LMICs, would it be reasonable to use existing evidence from HICs to argue that corporal punishment can undermine children's development in LMICs? There are reasons to believe that the “effects” of corporal punishment could be universal or context specific. On the one hand, it is likely that parental practices, including corporal punishment, may trigger different developmental consequences on children across countries and settings if the effects depend on the normativeness, prevalence, and severity of such practices within a given culture [10, 11]. Yet, it is also likely that corporal punishment might cause children to feel fear, anxiety, and sadness regardless of cultural factors, thus triggering emotional and physiological mechanisms that could lead to similar maladaptive outcomes across cultures and contexts [12, 13].

The purpose of this manuscript is to contribute to this conversation by discussing the main findings from the literature on the association between corporal punishment and child cognitive and social-emotional developmental outcomes in LMICs. Moreover, the study seeks to discuss the strength of evidence regarding issues of internal (i.e., causality, or whether it is corporal punishment that contributes

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to specific developmental outcomes or other ecological factors that relate to both corporal punishment and children's development) and external validity (i.e., generalizability, or the extent to which findings from one setting generalize to other populations or settings). To do so, I conducted a rapid review of the literature, which is a well-established approach that uses systematic methods to search and critically assess a body of literature in an accelerated timeframe [14]. Instead of combining results from multiple studies to estimate a summary average effect—as in meta-analyses,—rapid reviews seek to provide evidence on the quantity of evidence, overall quality on specific aspects (in the present study, issues of internal and external validity), and discuss future directions for research, policy, or practice [14, 15]. Using results from the review, I offer suggestions for future research aimed at examining the links between corporal punishment and child developmental outcomes in LMICs.

What is Corporal Punishment?

The UN Committee on the Rights of the Child (CRC) defines corporal punishment as “any punishment in which physical force is used and intended to cause some degree of pain or discomfort, however light” [5, para. 11]. Another definition, common in the literature, indicates that corporal punishment is “the use of physical force with the intention of causing a child to experience pain but not injury for the purposes of correction or control of the child's behavior” [16; p. 4]. Around the globe, corporal punishment usually comprises behaviors like spanking, slapping, pinching, kicking, and hitting children with hard objects like shoes or belts [1, 17].

While there is disagreement in the literature on whether there is a distinction between corporal punishment and abuse [e.g., 7, 18–20], drawing a line along the continuum of violence to distinguish abuse from punishment is not straightforward—particularly considering a global perspective—and is inconsistent with the CRC, which considers corporal punishment to be yet another form of abuse [5, 17, 21, 22]. Corporal punishment varies in terms of its severity, from punishment that is socially normative in several settings and does not cause apparent physical harm to the child (e.g., spanking) to punishment that is often considered as abusive and could cause evident physical harm (e.g., hitting with hard objects). Corporal punishment also varies on terms of its frequency, from occurring only once in the lifetime to being regular part of parents' “disciplinary” repertoire. While most researchers would agree that as corporal punishment is more severe or frequent it is “more abusive,” it is impossible to draw a meaningful line between both. For example, are three (or 10, or 20) spans a week a form of abuse, but fewer are not? Some researchers have argued that spans or slaps using an open hand is normative, whereas hitting

with objects is abusive [18]. Yet, this U.S.-based distinction between normative corporal punishment and abuse is far from universal and ignores that the severity and frequency of corporal punishment vary across cultures and settings, likely due to differences in local definitions and social norms on what is considered as appropriate discipline or punishment. For instance, while spanking is the most common and socially normative form of corporal punishment worldwide [17], in some LMICs hitting children with objects is more prevalent and socially normative than spanking [e.g., 23].

I followed the CRC and consider both corporal punishment and more severe abuse/physical maltreatment as forms of violence that are quantitatively but not qualitatively different (i.e., both represent different points in a continuum of violence). Furthermore, following the CRC and Straus' definitions, I conceptualize corporal punishment as punishment in which physical force is used to cause some degree of pain or discomfort, however light, for the purpose of controlling the child's behavior. Therefore, in the present review I do not consider studies in which children were exposed to physical violence for purposes other than managing their behavior, but restraint from making further conceptual distinctions between corporal punishment and abuse.

Conceptual Framework

This study considers a bioecological perspective [24], social learning theory [25], social information processing theory [26], attachment theory [27], and a neurobiological perspective [28, 29] to conceptualize the links between corporal punishment and child developmental outcomes. Overall, these theoretical models hypothesize that corporal punishment might interfere with cognitive and social-emotional development (as discussed below). As such, I focus on social-emotional and cognitive developmental outcomes, and not on motor/physical development nor non-developmental outcomes (e.g., risk of abuse). In this study, social-emotional development is conceptualized as a set of executive function (e.g., inhibitory control) and regulatory-related skills (e.g., emotional regulation) and behaviors (e.g., internalizing symptoms such as depression and externalizing behaviors like aggressiveness) [30]. Cognitive development comprises a set of sensorimotor skills, object relatedness, concept formation, memory, language and literacy, and academic achievement, among others [31].

The literature has traditionally used social learning, social-information processing, and attachment perspectives to argue that corporal punishment has the potential to produce maladaptive social-emotional outcomes [e.g., 19, 32]. In short, these perspectives suggest that by using corporal punishment parents (intentionally or unintentionally) model aggression to their children as a means to solve problems,

inculcate in the child expectations of aggression, and erode the attachment bond (by producing fear in the child). In time, processes of observational learning, a tendency to interpret others' behaviors as aggressive (i.e., a hostile attribution bias that leads to hypervigilance, heightened reactivity, and reactive aggression), and a weak attachment bond may spark social-emotional difficulties, mainly internalizing and externalizing behaviors. Indeed, meta-analyses show associations between a hostile attribution bias and externalizing problems like aggression in adolescence and adulthood [33] and between insecure attachment and internalizing problems like depression in childhood and adolescence [34].

While these mechanisms have been traditionally used to explain and study human development in cross-cultural contexts—explicitly or implicitly assuming universality,—there are ongoing discussions on the universality vs. specificity of processes of social learning and attachment [35–38]. Despite these conversations in the fields of developmental and cross-cultural psychology, researchers have used these U.S. based theoretical arguments to hypothesized links between corporal punishment and child developmental outcomes in LMICs, raising the question of whether corporal punishment might lead to similar outcomes or whether social-learning, social information, and attachment processes may be context-specific instead of universal.

More recently, emotional and physiological arousal has also been hypothesized to mediate the links between corporal punishment and negative cognitive and social-emotional outcomes [12, 13]. The argument, based on a cumulative risk perspective [29] and the dimensional model of adversity [39, 40], is that corporal punishment causes psychological and physiological arousal (e.g., pain, fear), which might elicit immediate compliance [19, 41]. Yet, corporal punishment causes pain and distress, involves threat of harm to the child, and the parent or caregiver—who should be a source of support and comfort—is the source of stress itself. Consequently, it could affect highly-plastic neural systems that underlie cognitive and social-emotional skills like emotion regulation, threat detection, and salience processing to facilitate rapid identification of threats in the environment, including the medial prefrontal cortex (PFC) and the hippocampus [12, 39]. From these perspectives, the consequences of corporal punishment scale in relation to the severity of threat involved, so that more severe or frequent punishment will cause consequences that are quantitatively larger in magnitude than those of “milder” forms of punishment, though qualitatively similar (i.e., in the same neural systems). Nascent empirical evidence supports these neurodevelopmental predictions, revealing structural differences in the PFC [42] and functional differences in the medial and lateral PFC [12] between children who were and were not exposed to corporal punishment.

However, does corporal punishment cause the emotional and physiological arousal posited by a neurodevelopmental perspective? Qualitative and quantitative evidence from Brazil [43], Colombia [44], Ghana [45], New Zealand [46], The Philippines [47], South Africa [48], and the United States [49], among other settings, indicate that children experience sadness, fear, pain, anger, shame, guilt and feel threatened when their parents employ corporal punishment. Moreover, children exposed to corporal punishment tend to exhibit high hormonal reactivity to stress [50], likely due to the physiological stress caused by the punishment. Considering that corporal punishment appears to produce emotional and physiological arousal across settings, it is likely that the consequences of corporal punishment might be more universal than context specific assuming a biologically based mechanism.

While these mechanisms indicate that corporal punishment might be detrimental for social-emotional and cognitive skills, some scholars have argued that these consequences might vary between children and across settings. First, from a developmental perspective, the age or developmental period when children are exposed to corporal punishment could lead to different consequences. Early childhood is a sensitive period of development when the stress response system and neural circuits are particularly plastic in response to experience [51, 52]. Therefore, it is likely that corporal punishment might exert a stronger influence on children's social-emotional and cognitive outcomes if it happens early in development. Second, variability in children's biological sensitivity to context [53] may make some children particularly susceptible to the (positive or negative) effects of corporal punishment. Indeed, prior studies found that genetic risk factors and the functioning of children's stress-response system moderate the association between corporal punishment and children's executive function and internalizing and externalizing problems [54, 55].

At the relational level, some have hypothesized that caregivers' warmth could moderate the links between corporal punishment and child outcomes, exerting a protective influence [56]. Yet, existing evidence is mixed, with some studies finding that maternal warmth or attachment bonds modify the association between corporal punishment and child developmental outcomes [56, 57] whereas other studies have failed to find moderation effects [58, 59]. Furthermore, these ideas and related empirical evidence [e.g., 59] are mostly based on a Western model of caregiving [36, 37], therefore making it difficult to understand similarities and differences in such dynamics in different settings.

At the contextual level, scholars [e.g., 10, 11] have argued that the extent to which corporal punishment is normative within a culture could potentially affect its severity—e.g., whether hitting children with objects is socially normative,—frequency—e.g., whether spanking children regularly

is adequate and even necessary—and the way children perceive it—e.g., as unjust or legitimate or as threatening or not threatening,—therefore potentially moderating its effects on child outcomes. For instance, a neurodevelopmental perspective indicates that the developmental consequences of corporal punishment will scale in relation to its severity or frequency [29, 39]. Similarly, it has been hypothesized that children who consider corporal punishment as “just” will be more likely to accept parents’ “disciplinary” message and possibly will not experience negative consequences [7, 60]. Yet, related empirical evidence is scarce and mixed, with some studies [61] showing that perceived justness does not moderate the association between the frequency of corporal punishment and externalizing problems.

Finally, assuming that corporal punishment is a stressor for children [44, 50], it is likely that it could lead to worse outcomes for children who are already exposed to other contextual stressors (e.g., poverty or violence) by leading to cumulative wear and tear of the allostatic systems [i.e., allostatic load; 62]. Given that adversities like extreme poverty and contextual violence are more prevalent in certain LMICs than in HICs [63–65], it is possible that children exposed to adversity in LMICs might be, on average, more susceptible to the potential developmental consequences of corporal punishment relative to their better-off peers.

These theoretical predictions indicate that corporal punishment could potentially lead to adverse outcomes through multiple pathways, but also that such consequences might vary between children and across settings. Despite these predictions, existing narrative reviews and meta-analyses of the links between corporal punishment and child outcomes have largely focused on samples from the U.S and other HICs and have failed to examine variation between children and settings. For example, only 30% of (i.e., 13) studies in Gershoff and Grogan-Kaylor's and 17% in Paolucci and Violato's meta-analysis considered samples from outside the U.S (not specifically from LMICs). In the most recent narrative review of prospective studies [4], only 3 out of the 69 studies included in the review employed samples from LMICs (i.e., Colombia, China, and Turkey). The underrepresentation of studies from LMICs is concerning considering that more than 90% of children in the world live in LMICs [66] and the results of studies from HICs might not generalize to LMICs.

This Study

Considering the limitations of prior literature reviews and the lack of systematized information about the links between corporal punishment and child developmental outcomes in LMICs, I conducted a rapid review of the literature on the association between corporal punishment and child developmental outcomes in LMICs, assessing the availability of

studies, patterns of findings, and strength of the evidence. Doing so, the review aims to inform future research on the developmental consequences of corporal punishment.

Method

I identified studies through three sources. First, I searched PsycInfo for studies published before October 2020. The searches used combinations of keywords and controlled vocabulary for corporal punishment (e.g., “child discipline,” “corporal punishment,” “punishment,” “spanking”) and child developmental outcomes (e.g., “child behavior,” “cognitive development,” “psychosocial development,” “childhood development”; see the Appendix for further details). Second, I examined all studies included in published meta-analyses on this topic [i.e., 2, 8, 19, 67–69]. Finally, I searched the International Bibliography of the Social Sciences using the same keywords with the goal of identifying more studies from LMICs. I used the World Bank Country and Lending Groups [70] to define the LMICs to be included in the review.

Types of Participants and Studies

The review considered for inclusion quantitative studies published between 2000 and October 2020 in peer-reviewed journals in English or Spanish examining the association between exposure to corporal punishment during early childhood, childhood, and adolescence and social-emotional and cognitive outcomes. I excluded studies using as main predictor endorsement of or attitudes towards corporal punishment (instead of actual use of corporal punishment) and retrospective studies without reference to the timing of exposure to corporal punishment (see Appendix for details on the inclusion and exclusion criteria and general search strategy).

Data Extraction

I extracted the following information: (1) list of authors; (2) year of publication; (3) country or countries included in the study; (4) age of participants; (5) sample size; (6) whether the sample was representative at the national or sub-national level; (7) measurement of corporal punishment; (8) who reported the variable on corporal punishment (e.g., caregiver or the child); (9) whether moderation analysis was conducted; (10) main outcome; (11) measurement of the outcome (e.g., report or direct assessment); (12) who reported the outcome; (13) analytic approach and method of estimation (e.g., ordinary least squares,

structural equation modeling, individual fixed-effects); (14) covariates included in the analyses; (15) main results; (16) results from moderation analyses. To assess overall findings, I coded whether each study identified any statistically significant associations between corporal punishment and social-emotional or cognitive outcomes.

To assess the strength of the evidence, I coded the studies according to the type of data used (cross-sectional or longitudinal) and method of estimation (e.g., linear model, fixed effects). I also identified whether each study employed any approach that goes beyond conventional covariate adjustment [71, 72], such as (1) controlling for prior levels of the outcome variable or (2) employing a “quasi-experimental” approach such as individual fixed effects or matching.

Results

A total of 192 studies met the inclusion criteria; only 42 (i.e., 22%) used samples from LMICs (Fig. 1). Table 1 presents characteristics of the 42 studies from LMICs. As evident in prior meta-analyses, research on the association between corporal punishment and child developmental outcomes in LMICs is limited relative to HICs. Yet, in the past 20 years

there has been progress in the number of publications using samples from LMICs (Fig. 2).

The studies comprised samples from 64 LMICs, six from East Asia and the Pacific, eight from Europe and Central Asia, 12 from Latin America and the Caribbean, nine from Middle East and North Africa, four from South Asia, and 25 from Sub-Saharan Africa (Fig. 3, Panel A). China was the most studied country (59% of or 26 studies), followed by Kenya (7 studies), Thailand (6), Colombia (5), and the Philippines (5), which were included in different studies using the same sample from the Parenting Across Cultures project [61, 73, 74].

Most studies [e.g., 74–76] assessed corporal punishment using adaptations of the Conflict Tactics Scale [77], inquiring about the frequency of any form of corporal punishment on a 7-point scale ranging from 0 (never) to 6 (frequent behavior). Other research asked caregivers about the usual tactics they use to manage children’s behaviors, including any form of corporal punishment [e.g., 78] or spanking in particular [e.g., 79]. Finally, other studies measured whether the child was ever exposed to any form of corporal punishment [e.g., 80].

Overall, the literature on the association between corporal punishment and social-emotional outcomes is robust

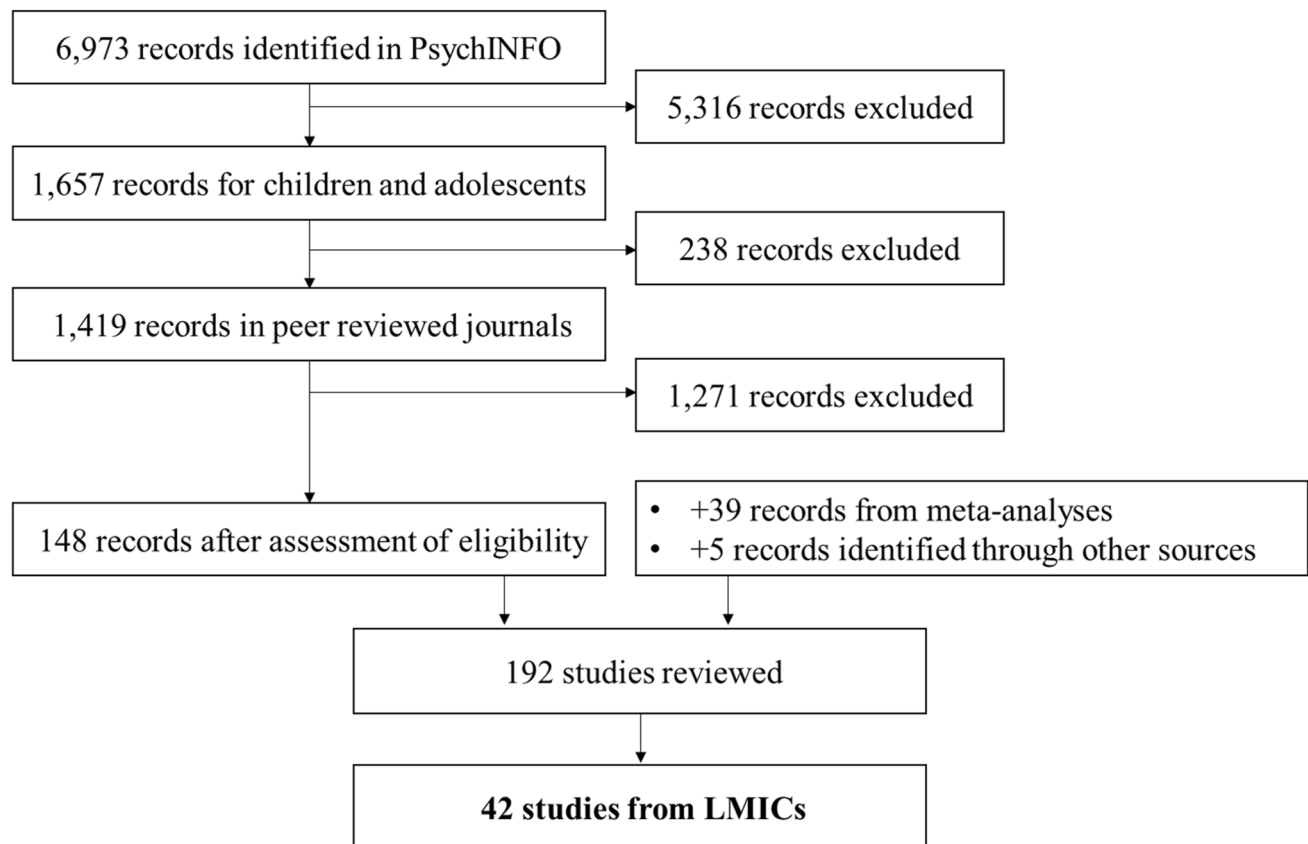


Fig. 1 Flow diagram of literature search

Table 1 Studies from LMICs of corporal punishment and child developmental outcomes: summary

Authors	Country	Data	N	Child age (in years) ^a	Measure of corporal punishment	Single informant for CP and outcome	Method	More than conventional covariate adjustment	Social-emotional	Cognitive	Moderation
1. Abolfotouh, El-Bourgy [87]	Egypt	CS	400	9–14	Frequency and severity of any CP	No	LM	No	A	NS	–
2. Alampay, Godwin [61]	6 LMICs	L	870	7–10	Frequency and severity of any CP	No ^d	SEM	No	A	–	NS—severity/justice
3. Alyabri and Goodman [85]	Yemen	CS	1166	7–10	Regular exposure to any CP	Yes	LM	No	A	A	–
4. Castell Ruiz, Frías Armenta [75]	Mexico	CS	170	6–12	Frequency of any CP	Yes	SEM	No	A	–	–
5. Cuartas, McCoy [80]	Colombia	L	1167	1–2	Ever exposed to any CP	No	LM, FE, PSM	Yes	–	A	–
6. Cuartas, Ward [84]	Colombia	CS	5801	4–16	Regular exposure to any CP	No ^d	PSM	Yes	A	A	–
7. Dede Yildirim and Roopnarine [83]	4 LMICs	CS	5628	3–5	Recent exposure to any CP	Yes	LM	No	–	A	–
8. Dede Yildirim, Roopnarine [78]	25 LMICs	CS	32,817	3–5	Recent exposure to any CP	Yes	LM	No	A	NS	NS—normativeness
9. Eisenberg, Chang [115]	China	CS	697	M = 8.3	Regular exposure to any CP	No ^d	SEM	No	A	–	–
10. Fu, Niu [94]	China	CS	675	9–13	Frequency of any CP	No ^d	LM	No	A	–	NS—warmth
11. Gershoff, Grogan-Kaylor [88]	5 LMICs	CS	292	8–12	Frequency of spanking	No ^d	LM	No	A	–	Yes—normativeness
12. Hecker, Hermeu [116]	Tanzania	CS	409	6–15	Ever exposed to any CP	Yes	LM	No	A	–	–
13. Hecker, Hermeu [86]	Tanzania	CS	409	6–15	Ever exposed to any CP	No ^d	SEM	No	A	A	–
14. Hesketh, Zheng [117]	China	CS	2203	7–13	Frequency of any CP	No ^d	LM	No	A	–	–
15. Khayyer [118]	Iran	CS	263	9–11	Frequency of any CP	Yes	MDT	No	A	–	–
16. Kohrt, Kohrt [81]	Mongolia	CS	99	3–10	Ever exposed to any CP	Yes	SEM	No	M	–	–
17. Lansford, Chang [90]	5 LMICs	CS	336	6–17	Frequency of any CP	No ^d	LM	No	A	–	Yes—normativeness

Table 1 (continued)

Authors	Country	Data	N	Child age (in years) ^a	Measure of corporal punishment	Single informant for CP and outcome	Method	More than conventional covariate adjustment	Social-emotional	Cognitive	Moderation
18. Lansford, Sharma [92]	6 LMICs	L	1196	7–10	Regular exposure to any CP	No ^d	SEM	No	A	–	Yes—warmth
19. Lau, Yu [105]	China	CS	3324	– ^b	Ever exposed to any CP	Yes	LM	No	A	–	–
20. Li, Shi [119]	China	CS	1222	2–6	Ever exposed to any CP	Yes	LM	No	A	–	–
21. Liu and Wang [89]	China	CS	1158	11–17	Frequency of any CP	Yes	LM	No	A	–	Yes—normativeness
22. Liu and Wang [55]	China	CS	158	10–12	Frequency of any CP	No	LM	No	A	–	Yes—HPA activity
23. Liu [120]	China	CS	2707	– ^c	Frequency of any CP	Yes	LM	No	A	–	–
24. Pace, Lee [79]	60 LMICs	CS	215,855	3–5	Recent exposure to spanking	Yes	LM	No	A	–	NS—normativeness
25. Roopnarine, Jin [95]	Guyana	CS	139	3–6	Severity of any corporal punishment	No	LM	No	A	–	NS—warmth
26. Salhi, Beatriz [82]	16 LMICs	CS	29,892	3–5	Recent exposure to any CP	Yes	LM	No	A	A	–
27. Schenek-Fontaine, Lansford [74]	5 LMICs	L	1418	8–12	Frequency of any CP	No ^d	LM, FE	Yes	A	–	–
28. Smith, Springer [121]	Jamaica	CS	563	13–19	Frequency of any CP	Yes	MANOVA	No	A	–	–
29. Steely and Rohner [104]	Jamaica	CS	97	7–18	Frequency and severity of any CP	Yes	LM	No	A	–	–
30. Wang and Liu [122]	China	L	702	6–9	Frequency of any CP	Yes	ALTM	Yes	A	–	–
31. Wang and Xing [123]	China	CS	761	7–15	Frequency of any CP	Yes	LM	No	A	–	–
32. Wang, Li [91]	China	CS	1164	11–16	Frequency of any CP	Yes	LM	No	A	–	Yes—normativeness
33. Wang, Wang [124]	China	CS	1971	7–12	Frequency of any CP	Yes	LM	No	A	–	–
34. Wang, Wang [93]	China	CS	667	8–13	Frequency of any CP	Yes	LM	No	A	–	Yes—warmth

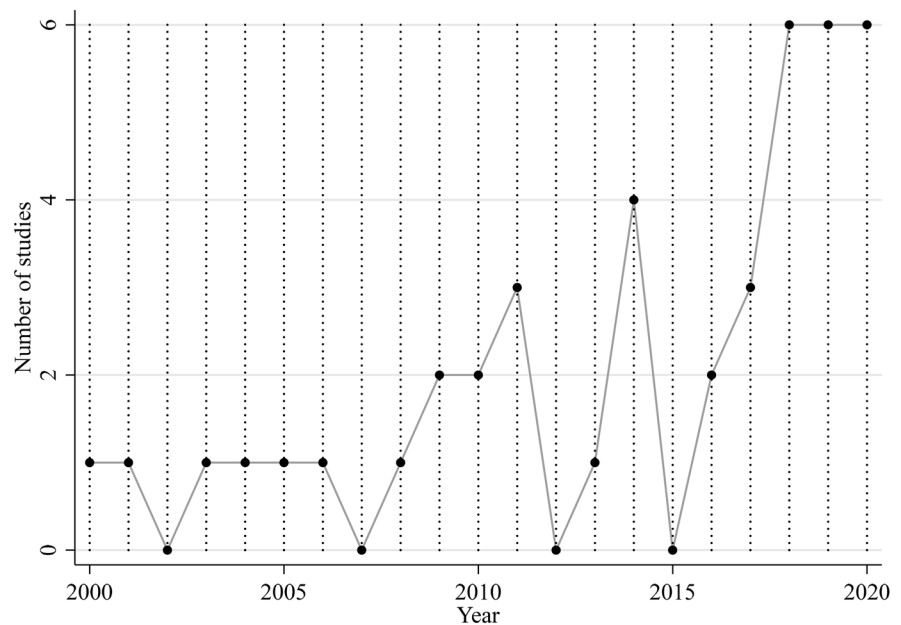
Table 1 (continued)

Authors	Country	Data	N	Child age (in years) ^a	Measure of corporal punishment	Single informant for CP and outcome	Method	More than conventional covariate adjustment	Social-emotional	Cognitive	Moderation
35. Xing and Wang [101]	China	L	454	M = 10.2	Frequency of any CP	Yes	CLPA	Yes	A	–	–
36. Xing and Wang [96]	China	CS	358	4–7	Frequency of any CP	Yes	LM	No	A	–	NS—warmth
37. Xing and Wang [98]	China	CS	150	9–11	Frequency of any CP	Yes	LM	No	A	–	Yes—HPA activity
38. Xing, Liu [102]	China	CS	273	M = 4.1	Frequency of any CP	No ^d	SEM	No	A	–	–
39. Xing, Yin [97]	China	L	223	M = 4.8	Frequency of any CP	No	LM	No	A	–	Yes—HPA activity
40. Xing, Wang [125]	China	CS	328	M = 4.1	Frequency of any CP	No ^d	SEM	No	A	–	NS—sex
41. Xing, Wang [99]	China	L	454	M = 10.2	Frequency of any CP	Yes	CLPA	Yes	A	–	Yes—sex
42. Zhu, Yu [100]	China	L	342	11–14	Frequency of any CP	Yes	CLPA	Yes	A	–	–

^aMean (M) reported when age range was not available^bNo clear information on age: the study comprised high school students of which 62% were below the age of 15^cNo clear information on age: the study comprised high school students^dDifferent informant (or direct assessment) for corporal punishment and some outcomes, but same informant for other outcomes

A adverse association between corporal punishment and outcome, NS statistically non-significant association, M mixed findings, CS cross-sectional, CP corporal punishment, L longitudinal, LM linear model, SEM structural equation modeling, FE individual fixed effects, PSM propensity score matching, MDT mean difference test, ALTM autoregressive latent trajectory model, CLPA cross-lagged panel analysis, HPA hypothalamic–pituitary–adrenal axis

Fig. 2 Trends in the number of studies examining the association between corporal punishment and child developmental outcomes in LMICs, 2000–2020



(both in terms of number of studies and in the consistency of findings), but evidence for cognitive outcomes is scarcer and more mixed. Importantly, there is no evidence that corporal punishment may relate to any positive developmental outcome in LMICs. A total of 40 (or 95% of) studies examined social-emotional outcomes, finding consistent positive associations between corporal punishment and social-emotional difficulties, including lower executive function and self-regulation (7 studies), higher internalizing (18 studies) and externalizing behaviors (25 studies), and other outcomes such as lower self-esteem (2 studies) or lower prosocial behaviors (1 study). Only one study from Mongolia [i.e., 81] failed to identify a statistically significant association between corporal punishment and social-emotional outcomes (i.e., internalizing symptoms), likely due to the small sample of 99 individuals.

Eight studies examined cognitive outcomes with mixed results. Corporal punishment was negatively related to young children's overall cognitive development in two studies covering 18 countries [80, 82], to language development in two studies covering three countries [83, 84], and to academic achievement in Tanzania (assessed using the result of a school exam) and Yemen (rated by teachers according to children's classroom and examination performance) [85, 86]. One study from Egypt [87] failed to identify significant associations between corporal punishment and academic achievement, and another study covering 25 African LMICs [78] failed to do so with literacy skills.

Even though there is evidence of variation in the association between corporal punishment and child developmental outcomes (i.e., moderation), in no case individual or contextual characteristics completely dissipated the links between

corporal punishment and negative developmental outcomes. A total of 16 studies examined moderation effects. Three studies [88–90] found that perceived normativeness partially buffered the negative association between corporal punishment and child outcomes, whereas one study found that it increased the observed association between corporal punishment and externalizing behaviors [91]. Two studies using large, representative samples from the Multiple Indicators Cluster Survey—MICS [78, 79] did not identify statistically significant moderation by perceived normativeness. Similarly, one study covering six countries [61] found that perceived justness of corporal punishment did not moderate the association between corporal punishment and child social-emotional outcomes.

In addition, one study [92] found that caregivers' warmth partially moderated the links between corporal punishment and internalizing outcomes if corporal punishment was infrequent but increased it if corporal punishment was frequent, whereas another study [93] found that caregivers' warmth (counterintuitively) increased the relation between corporal punishment and regulatory difficulties. Three studies did not find significant moderation effects for caregivers' warmth [94–96]. Three studies from China [55, 97, 98] found that the children's stress-response system functioning (as measured by cortisol stress reactivity) moderated the links between corporal punishment and social-emotional outcomes. Finally, two studies from China [99, 100] found contradictory results for whether sex moderates the association between corporal punishment and social-emotional outcomes.

While no study tested moderation by children's age, the reviewed studies indicate that corporal punishment might

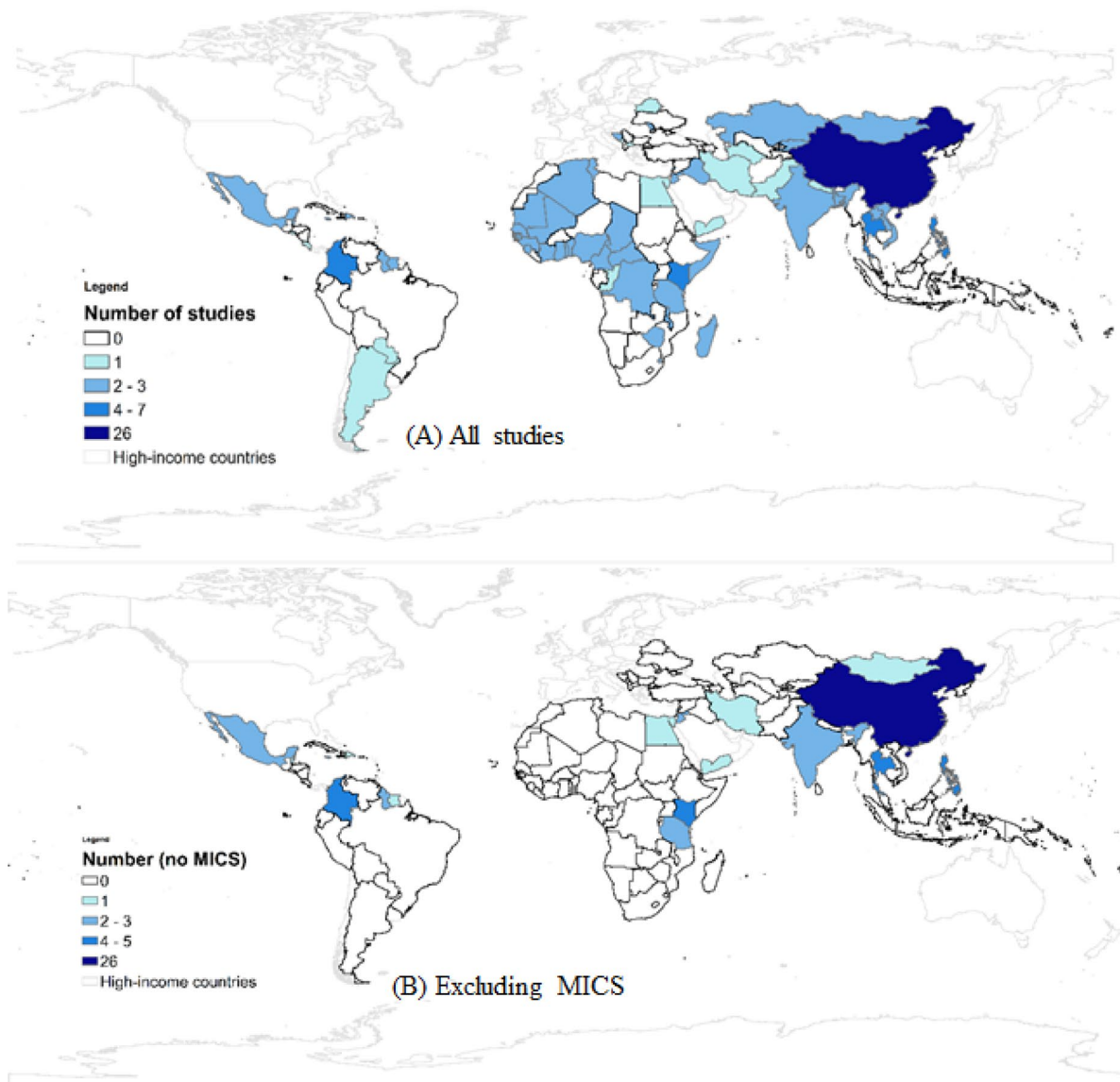


Fig. 3 Number of studies per country, 2000–2020

have a negative association with developmental outcomes throughout early childhood, childhood, and adolescence. Samples of children aged 0–6 years (early childhood) were included in 10 studies, 6–11 years (childhood) in 11 studies, 12–18 years (adolescence) in 10 studies, and the remaining 11 studies included multiple age groups. While the literature shows associations between corporal punishment and negative developmental outcomes for different age groups, the lack of moderation analyses makes it difficult to understand whether children might experience worse outcomes if corporal punishment takes place early in life as predicted by neurodevelopmental models of human development [51].

Despite the consistency of findings, issues of internal validity are common in the literature. First, only nine (20%)

out of the 42 studies used longitudinal data. Second, most studies (34 or 81%) relied on linear models or structural equation modeling with conventional covariate adjustment, suffering from potential issues of selection bias that hinder any causal claim. Finally, only seven studies (16%) used more than conventional covariate adjustment, including individual-level fixed effects [74, 80], matching methods [84], and autoregressive latent trajectory models or cross-lagged panel analysis [99–102]. In addition, more than half of studies relied on the same informant to measure corporal punishment and child developmental outcomes, which can produce upwardly biased coefficients and represent another threat to internal validity.

Similarly, there is limited evidence to determine the extent to which the negative association between corporal punishment and developmental outcomes is “universal” or context specific. First, 47 countries were covered in only three studies [78, 79, 82] that used nationally-representative but less precise data from the MICS’ Early Childhood Development Index (ECDI, which has been criticized for being a limited in scope measure with relatively poor psychometric properties [103]), so excluding those three studies leaves only 18 countries being represented in prior research (Fig. 3, Panel B). Similarly, besides the studies using data from the MICS, only one study [i.e., 84] used a nationally-representative sample, whereas the rest of studies used relatively small, non-representative samples that varied between 97 individuals in Jamaica [104] and 99 in Mongolia [81] to 3,324 in Chengdu, China [105]. Moreover, most countries have evidence from only one or two studies and four out of the seven studies that employed more than conventional covariate adjustment (see above) were conducted in China, two in Colombia, and one comprised small samples from five LMICs. As such, the coverage of studies with stronger internal validity in LMICs is limited.

Discussion

This study presented a rapid review of the literature examining the association between corporal punishment and child developmental outcomes in LMICs. In general, the literature shows associations between corporal punishment and negative social-emotional and cognitive outcomes and there is no evidence that corporal punishment might relate to any positive developmental outcome in LMICs. Despite the apparent consistency of findings, there are issues of internal and external validity that limit our understanding of the developmental effects of corporal punishment in LMICs, suggesting the need for additional research.

The research showing associations between corporal punishment and negative developmental outcomes in HICs has long been criticized for potential issues of causality [e.g., 7, 8]. On the one hand, there are concerns about reversed causality, i.e., it may be that children with more social-emotional or cognitive problems elicit more corporal punishment from their caregivers, and not the other way around. On the other hand, there are concerns about selection (or omitted variable) bias, as there may be unobserved variables (e.g., genetic factors, child temperament, caregiver’s depression) that simultaneously correlate with caregivers’ use of corporal punishment and children’s developmental outcomes, confounding the effects (and potentially overestimating the negative association between corporal punishment and developmental outcomes). Randomly assigning children to

“corporal punishment” or “no corporal punishment” conditions would be unethical. As such, researchers have relied on alternative approaches, including fixed effects and matching methods, leveraging longitudinal designs, to mitigate to some extent issues of selection bias and conclude that corporal punishment likely has a negative effect on children’s development and wellbeing [3, 4].

The present review showed that most studies from LMICs also employed conventional covariate adjustment, and only a handful used more conservative approaches to mitigate issues of selection bias. Despite these important efforts, even research employing more than conventional covariate adjustment relied on relatively strong identifying assumptions, thus being susceptible to bias. For instance, any omitted variable that varies in time and predicts both the use of corporal punishment and child developmental outcomes (e.g., caregivers’ depression, neighborhood crime, job or income loss) can bias the results from fixed effects models [71]. Similarly, matching relies on the (implausible) assumption of conditional independence, i.e., once the samples are matched, there are not remaining differences between children exposed to corporal punishment and not exposed to corporal punishment in unobserved confounders [106]. As such, even if researchers employ a long list of variables to match children exposed and not exposed to corporal punishment, any omitted confounder (e.g., maternal depression, child temperament, genetic factors) can bias the estimates. Despite this, only one study [i.e., 80] tested the sensitivity or robustness of estimates to alternative methods with different identifying assumptions, which is key to mitigate concerns about selection bias in observational data [107, 108]. These issues are aggravated by shared-method variance, common in the literature from LMICs, which can bias estimated coefficients and spuriously inflate the association between corporal punishment and developmental outcomes. In [summary](#), more research is needed to better understand whether corporal punishment in fact causes negative developmental outcomes in different LMICs.

Besides potential problems of internal validity, the literature from LMICs might suffer from issues of external validity. In particular, is the association between corporal punishment and child developmental outcomes universal or context-specific? and, for whom and when does corporal punishment produce negative developmental outcomes? Findings from this review suggest that there is a need of more replication studies using representative samples to strengthen the argument that corporal punishment might “universally” relate to detrimental outcomes across LMICs. Furthermore, there is scarce and rather mixed evidence on moderation, making it difficult to assess variation in the links between corporal punishment and developmental outcomes. For example, there are inconsistent findings on whether perceived normativeness or the relationship between caregivers

and children might buffer or exacerbate the links between corporal punishment and developmental outcomes. Similarly, it is empirically and conceptually unclear whether boys and girls might experience different consequences from corporal punishment. Besides, no study using samples from LMICs has tested whether there is a developmental period (e.g., early childhood) where children might be particularly sensitive to the effects of corporal punishment. In sum, little is known about variability at the individual and contextual level in the association between corporal punishment and developmental outcomes in LMICs.

Implications for Research

Studies using samples from LMICs reveal associations between corporal punishment and negative child developmental outcomes, in line with theoretical predictions [e.g., 12, 25, 26]. Yet, there is a need of more within and between country replications and studies with more conservative empirical strategies that mitigate issues of selection bias. In this section, I make suggestions for future research on the developmental consequences of corporal punishment in LMICs.

Establishing whether corporal punishment causes particular developmental outcomes is not straightforward. I argue so considering that experiments to identify the effect of corporal punishment are unfeasible and unethical, and it is almost impossible to find a source of exogenous variation in caregivers' use of corporal punishment to employ instrumental variables (IV) or regression discontinuity designs (RDD). For instance, interventions aimed at reducing corporal punishment often also encourage the use of other parenting behaviors (e.g., positive discipline) or might impact child development directly [see 109], thus making it endogenous and invalid as a potential instrument [71]. Even though some studies have used longitudinal data to guarantee that corporal punishment precedes child developmental outcomes, hence mitigating to some extent concerns about reversed causality, credibly alleviating issues of selection is way more difficult.

Because experiments and quasi-experimental designs with strong internal validity such as IV or RDD are unfeasible, what alternative approaches offer promise moving forward? I argue that future studies can leverage (1) approaches with weaker (i.e., more plausible) identifying assumptions and (2) sensitivity or robustness checks to test the plausibility of such assumptions. In particular, any causal claim relies upon an identifying assumption, and even experiments, often consider as the “gold standard” to identify causal effects, rely on the (plausible) assumption of equal potential outcomes between treatment and control groups [71]. In the case of corporal punishment, conventional covariate

adjustment is prone to significant threats to validity, as there may be several unobserved confounders (e.g., child temperament, shared genetic composition between children and parents, neighborhood characteristics) that make the conditional independence assumption unlikely. Therefore, leveraging longitudinal data with approaches such as individual fixed-effects, which relies on the (more plausible) assumption that there are no time-variant confounders left unexplained [71], or lagged-dependent variables, which reduces concerns about reversed-causality and selection [110], is warranted. Researchers have also used matching techniques (e.g., propensity score matching) to strengthen causal links between corporal punishment and child developmental outcomes [84, 111, 112]. Nonetheless, matching relies on the same assumption of conventional covariate adjustment (i.e., conditional independence), making it an useful tool to test balance in observed characteristics but not necessarily to credibly discard selection effects [113].

Given that the potential approaches to test causal links between corporal punishment and child outcomes rely on relatively strong identifying assumptions, future studies should provide more evidence on the sensitivity or robustness of their findings to make a credible case for causal effects. Indeed, the validity of prior research has been contested due to the infrequent use of sensitivity and robustness checks [8]. The robustness of results to multiple approaches with different identifying assumptions (e.g., lagged-dependent variables and individual fixed-effects) makes a causal link more credible [107]. Similarly, sensitivity tests to estimate the robustness or vulnerability of results to potential selection effects can be easily implemented and can constitute an important tool to strengthen causal inferences [108]. Finally, it is critical that future studies employ multi-informant designs and direct assessments of child developmental outcomes to reduce concerns about shared-method variance and potential bias caused by measurement issues.

Besides internal-validity issues, there is a need for more between study and across countries and cultures replications to address issues of limited external validity. There is vast variation between countries in the extent to which individuals consider corporal punishment to be normative and necessary to raise children [114]. Therefore, there is a need of more studies using samples from multiple countries that explicitly address the role of cultural factors in moderating the effects of corporal punishment on child outcomes. Likewise, future studies should consider a developmental perspective and examine whether the association between corporal punishment and child developmental outcomes varies according to children's age, perceived justness, children's relationship with their caregivers, markers of cumulative risk, and other developmental and contextual characteristics. Assessing moderation is critical to go beyond averages and

identify children who may be particularly vulnerable to the potential consequences of corporal punishment.

This review has limitations that inform future research. This was a rapid review of research indexed in a couple of psychology and social sciences databases or included in existing meta-analysis and systematic reviews. Therefore, several studies from LMICs indexed in medical or sociology databases (e.g., PubMed) were not included. Similarly, the rapid review did not offer evidence of the average association between corporal punishment and child developmental outcomes. Future researchers could conduct more comprehensive reviews of the literature, including multiple databases, and meta-analysis to understand the magnitude of the average association between corporal punishment and different child outcomes.

Summary

This review showed that the field has made important progress in understanding the developmental consequences of corporal punishment in LMICs. The studies from LMICs face the same challenges related to internal and external validity that have fueled the debate about the developmental consequences of corporal punishment in HICs [3, 8]. However, it is noteworthy that not a single study using samples from LMICs concluded that corporal punishment might be beneficial for children's cognitive or social-emotional skills under any circumstances. Quite the opposite, the research shows consistent associations between corporal punishment and negative child outcomes across settings. Following a precautionary principle, the lack of concluding evidence and ongoing debate must be settled in favor of prevention. Therefore, policymakers and practitioners should discourage the use of corporal punishment in LMICs.

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